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Candidate surname	Other names
USMAN	Grade A* Scholar
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Mentora Academy Internal Candidate Ref: MEN-EDX-2026-U4-W08	

Pearson Edexcel International Advanced Level

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

WEEK 8 ASSESSMENT: FACTORS AFFECTING K, $S_{\text{total}} = R \ln K$ & BRØNSTED-LOWRY THEORY

Assessment Paper:	WCH14/01 (Week 8 In-Depth Mastery)
Time Allowed:	2 hours 30 minutes
Total Marks:	150 Marks
Topics Examined:	Topic 13A.3: Factors Affecting K (Temperature)
	Topic 13A.4: Factors NOT Affecting K (Pressure, Concentration, Catalysts)
	Topic 13A.5: Relating Entropy to K ($S_{\text{total}} = R \ln K$ & van 't Hoff)
	Topic 14A.1: The Brønsted-Lowry Theory (Conjugate Pairs & Amphiprotic)

INSTRUCTIONS TO CANDIDATES

- Use black ink or ball-point pen. Fill in the boxes at the top of this page.
- Answer ALL questions in Section A, Section B, and Section C.
- In Section A (Questions 1 to 30), select your answer by placing a clear cross in the box [X].
- In Section B and C, show all working in calculation questions. Include appropriate units where required.
- Questions marked with an asterisk (*) are those where your quality of written communication and logical sequence of structure will be assessed.
- A Periodic Table and physical constants are printed on the back page of this booklet.

INFORMATION FOR CANDIDATES

- The total mark for this paper is 150.
- Section A: 30 marks; Section B: 90 marks; Section C: 30 marks.
- Full worked confidential teacher mark scheme is provided at the rear for post-assessment diagnostic.

SECTION A

Answer ALL questions. For each question, make a selection by putting a cross in one box [X]. If you change your mind, put a line through the box and then mark your new choice with a cross.

1 Which factor will change the numerical value of the equilibrium constant K_c for an endothermic reaction?

- ☐ A Increasing the total pressure
- ☐ B Increasing the concentration of reactants
- ☐ C Increasing the temperature
- ☐ D Adding a finely divided catalyst

(Total for Question 1 = 1 mark)

2 For the exothermic reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ ($H = -197 \text{ kJ mol}^{-1}$), how does an increase in temperature affect the equilibrium constant K_p and the position of equilibrium?

- ☐ A K_p increases; equilibrium shifts right
- ☐ B K_p decreases; equilibrium shifts left
- ☐ C K_p remains constant; equilibrium shifts left
- ☐ D K_p increases; equilibrium shifts left

(Total for Question 2 = 1 mark)

3 Which mathematical equation correctly relates the total entropy change, S_{total} , to the equilibrium constant K ?

- ☐ A $S_{\text{total}} = R \cdot \ln K$
- ☐ B $S_{\text{total}} = -R \cdot T \cdot \ln K$
- ☐ C $\ln K = R / S_{\text{total}}$
- ☐ D $K = R \cdot \ln(S_{\text{total}})$

(Total for Question 3 = 1 mark)

4 If a chemical reaction has an equilibrium constant $K = 1.0$ at 298 K, what is the value of S_{total} ?

- ☐ **A** $+8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ **B** $-8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ **C** $0 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ **D** $+298 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 4 = 1 mark)

5 A reaction has $S_{\text{total}} = +57.5 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K. What is the value of the equilibrium constant K ? ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- ☐ **A** 6.92
- ☐ **B** 1.01×10^3
- ☐ **C** 1.00
- ☐ **D** 4.82×10^{-4}

(Total for Question 5 = 1 mark)

6 According to the Brønsted-Lowry theory, an acid is defined as a species that:

- ☐ **A** Accepts a lone pair of electrons
- ☐ **B** Donates a proton (H^+ ion)
- ☐ **C** Produces OH^- ions in aqueous solution
- ☐ **D** Increases the oxidation state of hydrogen

(Total for Question 6 = 1 mark)

7 In the reaction: $\text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{SO}_4^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$, which pair represents a conjugate acid-base pair?

- ☐ A HSO_4^- and H_2O
- ☐ B HSO_4^- and SO_4^{2-}
- ☐ C H_2O and HSO_4^-
- ☐ D SO_4^{2-} and H_3O^+

(Total for Question 7 = 1 mark)

8 Which of the following chemical species is amphoteric (can act as both a Brønsted-Lowry acid and base)?

- ☐ A CO_3^{2-}
- ☐ B HCO_3^-
- ☐ C H_3O^+
- ☐ D Cl^-

(Total for Question 8 = 1 mark)

9 Why does increasing the total pressure have NO effect on the value of K_p for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$?

- ☐ A The mole fractions of all gases remain constant
- ☐ B K_p depends strictly on temperature through the thermodynamic relation $\Delta G_{\text{total}} = -RT \ln K$
- ☐ C The catalyst prevents pressure from affecting the equilibrium constant
- ☐ D The compressibility of nitrogen gas is negligible at high pressure

(Total for Question 9 = 1 mark)

10 In a graph of $\ln K_p$ against $1/T$ for an endothermic reaction ($H > 0$), what is the gradient of the straight line?

- ☐ A Positive gradient ($+H / R$)
- ☐ B Negative gradient ($-H / R$)
- ☐ C Zero gradient
- ☐ D Gradient equal to $-E_a / R$

(Total for Question 10 = 1 mark)

11 In liquid ammonia, self-ionisation occurs according to: $2\text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{NH}_2^-$. What is the conjugate base of NH_3 in this system?

- ☐ A NH_4^+
- ☐ B NH_2^-
- ☐ C N^{3-}
- ☐ D H^+

(Total for Question 11 = 1 mark)

12 When ethanoic acid is dissolved in pure liquid sulfuric acid: $\text{CH}_3\text{COOH} + \text{H}_2\text{SO}_4 \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{HSO}_4^-$. What role does CH_3COOH play?

- ☐ A Brønsted-Lowry acid
- ☐ B Brønsted-Lowry base
- ☐ C Dehydrating agent
- ☐ D Lewis acid

(Total for Question 12 = 1 mark)

13 For the equilibrium: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, the volume is suddenly halved at constant temperature. Immediately after the change, what is the value of the reaction quotient Q_p relative to K_p ?

- ☐ A $Q_p = K_p$
- ☐ B $Q_p = 2 K_p$
- ☐ C $Q_p = 0.5 K_p$
- ☐ D $Q_p = 4 K_p$

(Total for Question 13 = 1 mark)

14 A reaction has a very large positive equilibrium constant ($K = 1.0 \times 10^{12}$) at 298 K. What can be concluded about the reaction?

- ☐ A The reaction occurs at a very rapid rate at 298 K
- ☐ B The activation energy of the reaction must be zero
- ☐ C The position of equilibrium lies almost entirely on the side of the products
- ☐ D The reaction must be endothermic

(Total for Question 14 = 1 mark)

15 What is the conjugate acid of the dihydrogen phosphate ion, H_2PO_4^- ?

- ☐ A HPO_4^{2-}
- ☐ B PO_4^{3-}
- ☐ C H_3PO_4
- ☐ D H_4PO_4^+

(Total for Question 15 = 1 mark)

16 How does an increase in temperature affect an endothermic equilibrium in terms of entropy?

- ☐ A S_{system} increases
- ☐ B $S_{\text{surroundings}}$ ($-H/T$) becomes less negative, so S_{total} increases, increasing K
- ☐ C $S_{\text{surroundings}}$ becomes more negative, decreasing K
- ☐ D Both S_{system} and $S_{\text{surroundings}}$ remain constant

(Total for Question 16 = 1 mark)

17 For the Contact process: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $K_p = 3.0 \times 10^4 \text{ atm}^{-1}$ at 700 K. What is the value of S_{total} at 700 K? ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- ☐ A $+85.7 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $+28.8 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $-85.7 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+103.5 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 17 = 1 mark)

18 When sodium ethanoate is dissolved in water, the solution becomes slightly alkaline ($\text{pH} \approx 8.9$). Which equation represents the Brønsted-Lowry reaction responsible for this observation?

- ☐ A $\text{Na}^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NaOH}(\text{aq}) + \text{H}^+(\text{aq})$
- ☐ B $\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$
- ☐ C $\text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- ☐ D $2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$

(Total for Question 18 = 1 mark)

19 Why does the addition of a catalyst have NO effect on the value of S_{total} for a reaction?

- ☐ A A catalyst changes both H and S_{system} equally
- ☐ B State functions H , S_{system} , and $S_{\text{surroundings}}$ depend only on the initial and final states, not on the reaction pathway
- ☐ C Catalysts only function at absolute zero
- ☐ D A catalyst increases the entropy of the activation complex

(Total for Question 19 = 1 mark)

20 In the equilibrium: $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$, what is the role of $\text{H}_2\text{O}(\text{l})$?

- ☐ A Brønsted-Lowry acid
- ☐ B Brønsted-Lowry base
- ☐ C Conjugate acid of NH_4^+
- ☐ D Catalyst

(Total for Question 20 = 1 mark)

21 For a reaction with $H^\circ = -100 \text{ kJ mol}^{-1}$, which sketch of $\ln K$ against $1/T$ is correct?

- ☐ A A straight line with positive gradient ($+H^\circ/R$)
- ☐ B A straight line with negative gradient ($-H^\circ/R$)
- ☐ C A horizontal line parallel to the $1/T$ axis
- ☐ D An exponential curve concave upwards

(Total for Question 21 = 1 mark)

22 Which of the following species CANNOT act as a Brønsted-Lowry acid?

- ☐ A NH_4^+
- ☐ B HSO_3^-
- ☐ C BF_3
- ☐ D H_2O

(Total for Question 22 = 1 mark)

23 When nitric acid dissolves in concentrated sulfuric acid, the nitronium ion is formed: $\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightleftharpoons \text{NO}_2^+ + \text{H}_3\text{O}^+ + 2\text{HSO}_4^-$. Which species acts as a Brønsted-Lowry base in this reaction?

- ☐ A H_2SO_4
- ☐ B HNO_3
- ☐ C NO_2^+
- ☐ D HSO_4^-

(Total for Question 23 = 1 mark)

24 If S_{total} for a reaction is negative at a particular temperature, what is the value of K ?

- ☐ A $K < 0$
- ☐ B $0 < K < 1$
- ☐ C $K = 1$
- ☐ D $K > 1$

(Total for Question 24 = 1 mark)

25 Consider the water-gas shift reaction: $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$ ($\Delta H^\circ = -41.2 \text{ kJ mol}^{-1}$). How will the equilibrium yield of H_2 and the value of K_c change if the temperature is lowered?

- ☐ A Yield of H_2 increases; K_c increases
- ☐ B Yield of H_2 decreases; K_c decreases
- ☐ C Yield of H_2 increases; K_c remains constant
- ☐ D Yield of H_2 decreases; K_c increases

(Total for Question 25 = 1 mark)

26 Which of the following represents the conjugate base of the ammonium ion, NH_4^+ ?

- ☐ A NH_3
- ☐ B NH_2^-
- ☐ C NH_2OH
- ☐ D N_2

(Total for Question 26 = 1 mark)

27 At 500 K, K_p for a reaction is 10.0 atm. If the total pressure is doubled by compression at 500 K, what is the new value of K_p ?

- ☐ A 20.0 atm
- ☐ B 5.0 atm
- ☐ C 10.0 atm
- ☐ D 100.0 atm

(Total for Question 27 = 1 mark)

28 Why is the hydrogen carbonate ion, HCO_3^- , classified as amphoteric?

- ☐ A It can donate H^+ to form CO_3^{2-} and accept H^+ to form H_2CO_3
- ☐ B It can form both ionic and covalent bonds
- ☐ C It dissolves in both polar and non-polar solvents
- ☐ D It can be oxidised to CO_2 and reduced to C

(Total for Question 28 = 1 mark)

29 A reaction has $K = 2.5 \times 10^{-5}$ at 298 K. What is the value of S_{total} at 298 K? ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- ☐ A $+88.1 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ B $-88.1 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ C $-10.6 \text{ J K}^{-1} \text{ mol}^{-1}$
- ☐ D $+10.6 \text{ J K}^{-1} \text{ mol}^{-1}$

(Total for Question 29 = 1 mark)

30 In the reaction: $\text{CH}_3\text{COOH} + \text{HClO}_4 \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{ClO}_4^-$, which species is the conjugate base?

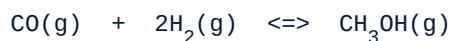
- ☐ A CH_3COOH
- ☐ B HClO_4
- ☐ C $\text{CH}_3\text{COOH}_2^+$
- ☐ D ClO_4^-

(Total for Question 30 = 1 mark)

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

31 Methanol is produced industrially from synthesis gas (syngas) according to the reversible reaction:

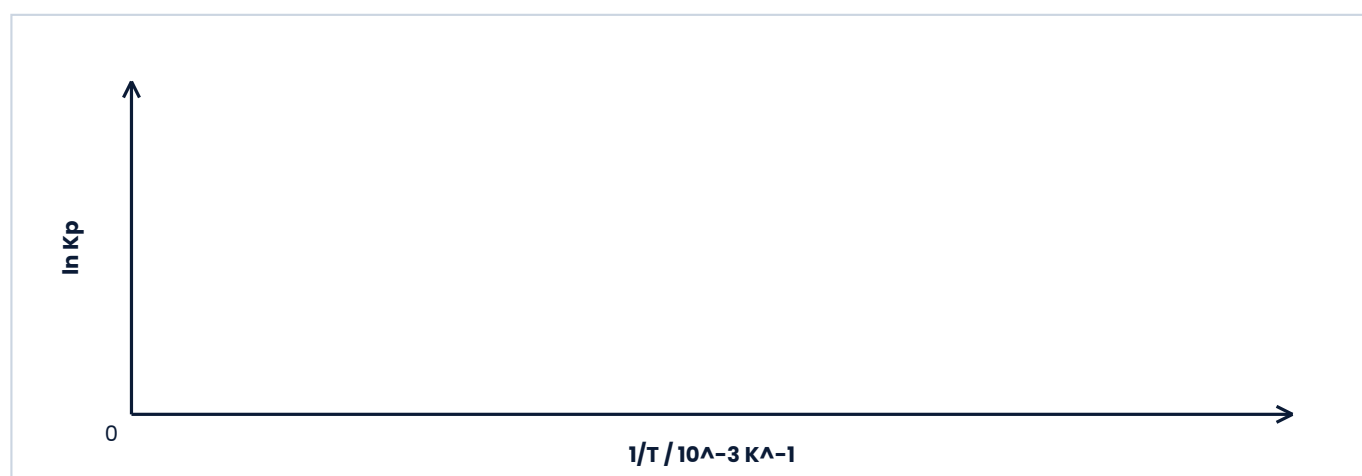


The equilibrium constant, K_p , was measured experimentally at several temperatures. The experimental data are tabulated below:

Temperature T / K	500	550	600	650	700
$1/T / 10^{-3} \text{ K}^{-1}$	2.00	1.82	1.67	1.54	1.43
K_p / kPa^{-2}	6.23×10^{-3}	3.81×10^{-4}	3.42×10^{-5}	4.32×10^{-6}	7.14×10^{-7}
$\ln K_p$	-5.08	-7.87	-10.28	-12.35	-14.15

(a) In the space below, sketch the graph of $\ln K_p$ against $1/T$.

(2)



(b) The van 't Hoff equation states that:

$$\ln K_p = (-H^\circ / R) \times (1/T) + (S^\circ_{\text{system}} / R)$$

Calculate the gradient of the line using the data points at $T = 500 \text{ K}$ and $T = 700 \text{ K}$. Include the sign and units of the gradient.

(3)

Gradient = units

(c) Using your gradient from (b), calculate the standard enthalpy change, H° , for the reaction in kJ mol^{-1} . Include a sign in your answer. ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$).

(3)

$H^\circ = \dots\dots\dots \text{kJ mol}^{-1}$

(d) Use your value of H° and the experimental value of $\ln K_p$ at 500 K to calculate the standard entropy change of the system, S°_{system} , in $\text{J K}^{-1} \text{ mol}^{-1}$.

(4)

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(e) Explain why the standard entropy change of the system, S°_{system} , is negative for this reaction, and why the value of K_p decreases so steeply as temperature increases.

(6)

(Total for Question 31 = 18 marks)

32 A common misconception among chemistry students is that changing the pressure alters the value of the equilibrium constant K_p .

Consider the gas-phase dissociation:



At 298 K, this system is at dynamic equilibrium inside a gas syringe. The equilibrium partial pressures are:

- $p(\text{N}_2\text{O}_4) = 0.800 \text{ atm}$
- $p(\text{NO}_2) = 0.400 \text{ atm}$

(a) Calculate the value of K_p at 298 K, including units. (3)

$K_p = \dots\dots\dots \text{units} \dots\dots\dots$

(b) The plunger of the syringe is suddenly pushed in, halving the volume of the gas mixture at constant temperature.

(i) State the instantaneous partial pressure of each gas immediately after the volume is halved (before any reaction occurs).

(ii) Calculate the instantaneous reaction quotient, Q_p , immediately after compression. (4)

$Q_p = \dots\dots\dots \text{atm}$

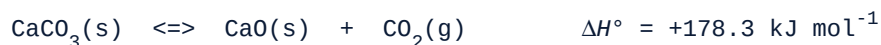
(c) Compare your value of Q_p from (b) with the equilibrium constant K_p from (a). Deduce, with mathematical reasoning, in which direction the reaction must proceed to re-establish dynamic equilibrium. **(4)**

(d) Explain clearly why the position of equilibrium shifted when pressure was increased, even though the numerical value of K_p remained completely unchanged. **(4)**

(e) If the same compression experiment is repeated for the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, explain why halving the volume does NOT cause any shift in the position of equilibrium. **(3)**

(Total for Question 32 = 18 marks)

33 The thermal decomposition of calcium carbonate in an industrial lime kiln is represented by:



Thermodynamic data at 298 K are shown below:

- $S^\circ[\text{CaCO}_3(\text{s})] = 92.9 \text{ J K}^{-1} \text{ mol}^{-1}$
- $S^\circ[\text{CaO}(\text{s})] = 38.2 \text{ J K}^{-1} \text{ mol}^{-1}$
- $S^\circ[\text{CO}_2(\text{g})] = 213.6 \text{ J K}^{-1} \text{ mol}^{-1}$

(a) Calculate the standard entropy change of the system, S°_{system} , in $\text{J K}^{-1} \text{ mol}^{-1}$. (2)

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(b) Calculate $S_{\text{surroundings}}$ and S_{total} at 298 K.

Using the relationship $S_{\text{total}} = R \ln K$, calculate the equilibrium constant K_p at 298 K. ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$). (5)

$S_{\text{total}} (298 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$ $K_p (298 \text{ K}) = \dots\dots\dots \text{atm}$

(c) Calculate S_{total} and K_p at 1200 K (approx 930 °C), assuming H° and S°_{system} do not vary significantly with temperature.

(4)

$S_{\text{total}} (1200 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$ $K_p (1200 \text{ K}) = \dots\dots\dots \text{atm}$

(d) For this heterogeneous reaction, write the expression for K_p in terms of partial pressures. Using your value of K_p at 1200 K from (c), state the equilibrium partial pressure of carbon dioxide inside a sealed kiln at 1200 K.

(3)

$p(\text{CO}_2) = \dots\dots\dots \text{atm}$

(e) In commercial lime kilns, air is continuously blown through the kiln to sweep away the carbon dioxide gas.

Explain how this engineering measure ensures 100 % conversion of limestone to quicklime at temperatures lower than 1200 K.

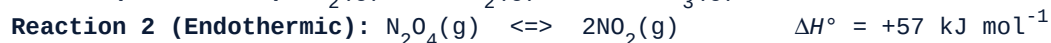
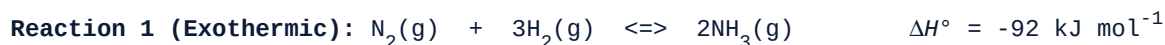
(4)

(Total for Question 33 = 18 marks)

*34 Two approaches are used to predict the effect of conditions on reversible reactions:

- Qualitative approach: Le Chatelier's Principle
- Quantitative thermodynamic approach: The Second Law of Thermodynamics ($S_{\text{total}} = R \ln K$)

Consider the two industrial reactions:



(a) For Reaction 1, explain using Le Chatelier's principle how an increase in temperature affects the equilibrium yield of ammonia.

(3)

*(b) Evaluate how the thermodynamic equation:

$$S_{\text{total}} = S_{\text{system}} - (H / T) = R \ln K$$

provides a fundamental explanation for the effect of temperature and catalysts on equilibrium that Le Chatelier's principle cannot provide.

In your answer, you should include:

- an explanation of how an increase in temperature changes $S_{\text{surroundings}}$ for both exothermic and endothermic reactions
- a deduction of how this temperature change affects S_{total} and consequently the equilibrium constant K
- a rigorous thermodynamic explanation of why adding a catalyst changes the rate of reaction but CANNOT change the value of K or the equilibrium yield.

(6)

(c) In Reaction 2, the brown gas NO_2 is produced from colorless N_2O_4 .

A sealed syringe containing an equilibrium mixture at 20°C is placed into hot water at 80°C .

Describe the color change observed and explain the thermodynamic driving force for this change.

(4)

(d) Chemical engineers designing Haber process plants use an operating temperature of approx 450°C (723 K).

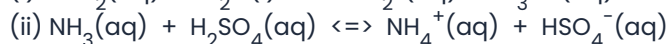
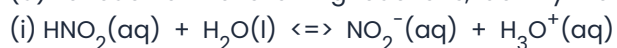
Explain the economic and thermodynamic rationale for selecting this temperature, resolving the conflict between equilibrium yield (thermodynamics) and reaction velocity (kinetics).

(5)

(Total for Question 34 = 18 marks)

35 The Brønsted–Lowry concept provides a comprehensive framework for understanding proton transfer reactions in both aqueous and non-aqueous media.

(a) For each of the following reactions, identify the two Brønsted–Lowry conjugate acid–base pairs:



(4)

(b) The hydrogencarbonate ion, HCO_3^- , is amphoteric.

(i) Write an equation showing HCO_3^- acting as a Brønsted–Lowry acid in water.

(ii) Write an equation showing HCO_3^- acting as a Brønsted–Lowry base in water.

(4)

(c) In pure liquid ammonia, ammonium chloride, NH_4Cl , behaves as an acid, while sodium amide, NaNH_2 , behaves as a base.

(i) Write the autoionisation equilibrium equation for liquid ammonia.

(ii) Write an ionic equation for the neutralisation reaction between NH_4^+ and NH_2^- in liquid ammonia.

(4)

(d) When pure nitric acid is dissolved in concentrated sulfuric acid (as in the preparation of nitrating mixtures for benzene electrophilic substitution), the following equilibrium is established:



Identify which reactant acts as a Brønsted-Lowry base and explain why.

(3)

(e) Water self-ionises according to: $2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$ ($H^\circ = +57.0 \text{ kJ mol}^{-1}$).

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

At 373 K (100 °C), $K_w = 5.13 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$.

Calculate the pH of pure neutral water at 100 °C. Explain why pure water at 100 °C is still strictly neutral despite having a pH below 7.0.

(3)

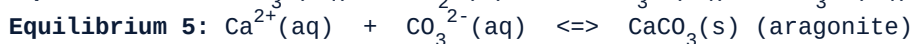
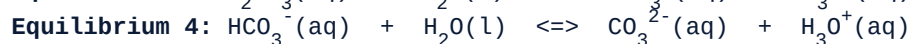
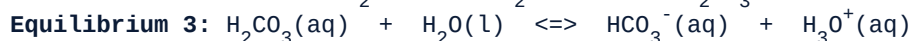
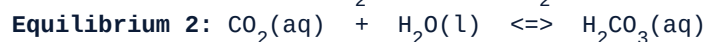
pH of water at 100 °C =

(Total for Question 35 = 18 marks)

SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 The absorption of anthropogenic carbon dioxide by the world's oceans drives ocean acidification through a cascade of chemical equilibria:



(a) Since the Industrial Revolution, atmospheric $p(\text{CO}_2)$ has risen from 280 ppm (2.80×10^{-4} atm) to 420 ppm (4.20×10^{-4} atm).

Using Le Chatelier's principle, explain how this 50 % increase in atmospheric CO_2 shifts Equilibria 1, 2, and 3, and state the effect on oceanic $[\text{H}_3\text{O}^+]$ and pH.

(4)

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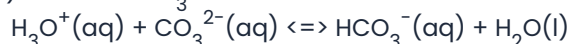
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(b) The added H_3O^+ ions react with carbonate ions according to the net neutralization:



Explain why ocean acidification causes the concentration of dissolved carbonate ions, $[\text{CO}_3^{2-}(\text{aq})]$, to DECREASE rather than increase.

(3)

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(c) Marine corals and molluscs build exoskeletons of aragonite, $\text{CaCO}_3(\text{s})$. The solubility product of aragonite is $K_{\text{sp}} = 6.0 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$.

The saturation state, Ω , is defined as: $\Omega = [\text{Ca}^{2+}][\text{CO}_3^{2-}] / K_{\text{sp}}$

In pre-industrial surface seawater, $[\text{Ca}^{2+}] = 0.0102 \text{ mol dm}^{-3}$ and $[\text{CO}_3^{2-}] = 2.40 \times 10^{-4} \text{ mol dm}^{-3}$.

(i) Calculate the pre-industrial saturation state, Ω .

(ii) If ocean acidification lowers $[\text{CO}_3^{2-}]$ to $5.0 \times 10^{-7} \text{ mol dm}^{-3}$, calculate the new value of Ω and predict whether coral skeletons will form or dissolve.

(5)

(d) Suggest one ecological and one economic consequence of the collapse of marine calcifying ecosystems.

(3)

(Total for Question 36 = 15 marks)

37 In an industrial Haber ammonia synthesis loop:



(a) Calculate S_{total} and K_p at 298 K (25 °C). ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$). **(4)**

$S_{\text{total}} (298 \text{ K}) = \dots\dots\dots \text{ J K}^{-1} \text{ mol}^{-1}$ $K_p (298 \text{ K}) = \dots\dots\dots \text{ atm}^{-2}$

(b) Calculate S_{total} and K_p at 723 K (450 °C), the typical operating temperature of the reactor.
Assume H° and S°_{system} remain constant. **(4)**

$S_{\text{total}} (723 \text{ K}) = \dots\dots\dots \text{ J K}^{-1} \text{ mol}^{-1}$ $K_p (723 \text{ K}) = \dots\dots\dots \text{ atm}^{-2}$

(c) Compare your values of K_p at 298 K and 723 K.

Explain why the industrial synthesis is operated at 450 °C, where K_p is roughly ten orders of magnitude smaller than at room temperature.

(4)

(d) Under industrial conditions (450 °C, 200 atm), only approximately 15 % of the feed gas is converted to ammonia on a single pass through the catalytic reactor.

Describe the industrial engineering loop used to achieve an overall 98 % conversion of the nitrogen and hydrogen feed gases, explaining how ammonia is separated from unreacted gases.

(3)

(Total for Question 37 = 15 marks)

DATA SHEET: PERIODIC TABLE OF THE ELEMENTS & PHYSICAL CONSTANTS

Physical Constant	Symbol	Numerical Value & Units
Gas constant	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	L or N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Ionic product of water (298 K)	K_w	$1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
Specific heat capacity of water	c	$4.18 \text{ J g}^{-1} \text{ K}^{-1} = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$
Standard pressure	p°	$1.00 \times 10^5 \text{ Pa} = 100 \text{ kPa} = 1.00 \text{ bar}$
Standard temperature	T°	$298.15 \text{ K} = 25.0 \text{ }^\circ\text{C}$

Group →	1	2	3	4	5	6	7	0 / 8
Period 1	H 1.0							He 4.0
Period 2	Li 6.9	Be 9.0	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0	Ne 20.2
Period 3	Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5	Ar 39.9
Period 4	K 39.1	Ca 40.1	Ga 69.7	Ge 72.6	As 74.9	Se 79.0	Br 79.9	Kr 83.8
Period 5	Rb 85.5	Sr 87.6	In 114.8	Sn 118.7	Sb 121.8	Te 127.6	I 126.9	Xe 131.3
Period 6	Cs 132.9	Ba 137.3	Tl 204.4	Pb 207.2	Bi 209.0	Po [209]	At [210]	Rn [222]

CONFIDENTIAL TEACHER MARK SCHEME & EXAMINER TRAP COMMENTARY

Pearson Edexcel International Advanced Level Chemistry – Unit 4 (WCH14/01) – Week 8 Assessment (150 Marks)

Q	Correct Answer	Key Concept & Examiner Guidance	Mark
1	C	Only temperature changes the numerical value of equilibrium constants (K_c and K_p).	1
2	B	For exothermic reactions, increasing T decreases K_p and shifts equilibrium to the left.	1
3	A	Fundamental relation: $S_{\text{total}} = R \ln K \Rightarrow \ln K = S_{\text{total}} / R$.	1
4	C	When $K = 1.0$, $\ln K = 0$, so $S_{\text{total}} = R \cdot 0 = 0 \text{ J K}^{-1} \text{ mol}^{-1}$.	1
5	B	$\ln K = 57.5 / 8.314 = 6.916 \Rightarrow K = e^{(6.916)} = 1.008 \times 10^3 \approx 1.01 \times 10^3$.	1
6	B	Brønsted-Lowry acid is a proton (H^+) donor; base is a proton acceptor.	1
7	B	HSO_4^- and SO_4^{2-} differ by exactly one proton (H^+), forming a conjugate acid-base pair.	1
8	B	HCO_3^- can donate H^+ to form CO_3^{2-} and accept H^+ to form H_2CO_3 , so it is amphoteric.	1
9	B	K_p depends strictly on temperature via $S_{\text{total}} = R \ln K$; pressure alters position, not K_p .	1
10	B	van 't Hoff plot: slope = $-\text{H}^\circ / R$; for $\text{H}^\circ > 0$, the slope is negative.	1
11	B	$\text{In } 2\text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{NH}_2^-$, NH_3 loses a proton to form NH_2^- (conjugate base).	1
12	B	CH_3COOH accepts a proton from H_2SO_4 to become $\text{CH}_3\text{COOH}_2^+$, acting as a Brønsted base.	1
13	B	Halving V doubles all partial pressures: $Q_p = (2p_{\text{NO}_2})^2 / (2p_{\text{N}_2\text{O}_4}) = 4/2 \cdot K_p = 2 K_p$.	1
14	C	A massive K (10^{12}) indicates almost 100% conversion to products at equilibrium.	1
15	C	Conjugate acid forms by adding one proton (H^+) to H_2PO_4^- , yielding H_3PO_4 .	1
16	B	For endothermic ($\text{H} > 0$), $S_{\text{surr}} = -\text{H}/T$ is negative; higher T makes it less negative, increasing S_{total} and K.	1
17	A	$S_{\text{total}} = R \ln K_p = 8.314 \cdot \ln(3.0 \times 10^4) = 8.314 \cdot 10.309 = +85.7 \text{ J K}^{-1} \text{ mol}^{-1}$.	1
18	B	CH_3COO^- accepts a proton from water to form CH_3COOH and OH^- (hydrolysis/alkaline).	1
19	B	Thermodynamic properties (state functions) depend only on initial and final states, not path.	1
20	A	Water donates a proton to NH_3 , acting as a Brønsted-Lowry acid.	1
21	A	Slope = $-\text{H}^\circ / R$; when H° is negative (-100 kJ), $-(-100)/R$ is positive, so slope is positive.	1
22	C	BF_3 has no protons to donate (it is a Lewis acid, not a Brønsted-Lowry acid).	1
23	B	HNO_3 accepts a proton from H_2SO_4 to form H_2NO_3^+ (which loses water to form NO_2^+), so HNO_3 acts as a base.	1
24	B	When $S_{\text{total}} < 0$, $\ln K = S_{\text{total}} / R$ is negative, so $0 < K < 1$ (equilibrium favours reactants).	1
25	A	Exothermic reaction: lowering T increases K_c and shifts equilibrium forward (higher H_2 yield).	1
26	A	NH_4^+ loses a proton to form ammonia, NH_3 .	1
27	C	K_p depends ONLY on temperature; doubling pressure leaves K_p unchanged at 10.0 atm.	1
28	A	Definition of amphoteric: can both donate H^+ and accept H^+ .	1
29	B	$S_{\text{total}} = R \ln K = 8.314 \cdot \ln(2.5 \times 10^{-5}) = 8.314 \cdot (-10.597) = -88.1 \text{ J K}^{-1} \text{ mol}^{-1}$.	1
30	D	HClO_4 donates a proton to become ClO_4^- (its conjugate base).	1

Question 31: Van 't Hoff Plot: Methanol Synthesis (18 Marks)

(a) [2 Marks]

- Axes labeled $\ln K_p$ on y-axis, $1/T / 10^{-3} \text{ K}^{-1}$ on x-axis [1]
- Straight line with positive slope (rising from left to right as $1/T$ increases) [1].

(b) [3 Marks]

- Gradient = $(\ln K_p) / (1/T) = (-5.08 - (-14.15)) / [(2.00 - 1.43) \times 10^{-3} \text{ K}^{-1}]$ [1]
- = $(+9.07) / (0.57 \times 10^{-3} \text{ K}^{-1}) = +15\,912 \text{ K}$ (accept $+15\,800$ to $+16\,100 \text{ K}$) [1]
- Sign is POSITIVE; units are Kelvin (K) [1].

(c) [3 Marks]

- Gradient = $-H^\circ / R \Rightarrow H^\circ = -\text{Gradient} \times R$ [1]
- $H^\circ = -(+15\,912 \text{ K}) \times (8.314 \text{ J mol}^{-1} \text{ K}^{-1}) = -132\,300 \text{ J mol}^{-1}$ [1]
- = $-132.3 \text{ kJ mol}^{-1}$ (must include minus sign; accept -131 to -134 kJ mol^{-1}) [1].

(d) [4 Marks]

- At 500 K: $1/T = 2.00 \times 10^{-3} \text{ K}^{-1}$; $\ln K_p = -5.08$ [1]
- $-5.08 = (-H^\circ / R) \times (1/T) + (S^\circ_{\text{sys}} / R) = (15\,912 \times 2.00 \times 10^{-3}) + (S^\circ_{\text{sys}} / 8.314)$ [1]
- $-5.08 = 31.824 + (S^\circ_{\text{sys}} / 8.314) \Rightarrow S^\circ_{\text{sys}} / 8.314 = -5.08 - 31.824 = -36.904$ [1]
- $S^\circ_{\text{sys}} = -36.904 \times 8.314 = -306.8 \text{ J K}^{-1} \text{ mol}^{-1}$ (accept -305 to $-310 \text{ J K}^{-1} \text{ mol}^{-1}$) [1].

(e) [6 Marks]

- 3 moles of gas reactants ($1 \text{ CO} + 2 \text{ H}_2$) produce 1 mole of gas product (CH_3OH) [2]
- Substantial loss of translational disorder / fewer microstates makes S°_{system} negative [1]
- $S_{\text{total}} = S_{\text{system}} - (H^\circ / T)$. Because H° is negative, $-H^\circ/T$ is positive ($S_{\text{surr}} > 0$) [1]
- As T increases, S_{surr} decreases rapidly, making S_{total} less positive (more negative) [1]
- Because $\ln K_p = S_{\text{total}} / R$, K_p decreases exponentially with increasing temperature [1].

Question 32: Mathematical Proof: Pressure Does Not Alter K_p (18 Marks)

(a) [3 Marks]

- $K_p = p(\text{NO}_2)^2 / p(\text{N}_2\text{O}_4)$ [1]
- $K_p = (0.400)^2 / (0.800) = 0.160 / 0.800 = 0.200$ [1]
- Units: atm (or 20.3 kPa if converted) [1].

(b) [4 Marks]

- (i) Halving volume doubles all concentrations/pressures: $p(\text{N}_2\text{O}_4) = 1.600 \text{ atm}$; $p(\text{NO}_2) = 0.800 \text{ atm}$ [2]
- (ii) $Q_p = (p_{\text{NO}_2})^2 / p_{\text{N}_2\text{O}_4} = (0.800)^2 / (1.600) = 0.640 / 1.600 = 0.400 \text{ atm}$ [2].

(c) [4 Marks]

- $Q_p = 0.400 \text{ atm}$, which is greater than $K_p = 0.200 \text{ atm}$ ($Q_p = 2 K_p$) [1]
- Because $Q_p > K_p$, the ratio of products to reactants is too high for equilibrium [1]
- The system must shift to consume NO_2 and produce N_2O_4 (shift to the LEFT) [1]
- This decreases numerator and increases denominator until Q_p returns to $0.200 \text{ atm} = K_p$ [1].

(d) [4 Marks]

- K_p is a thermodynamic constant that depends solely on temperature: $K_p = \exp(S_{\text{total}} / R)$ [1]
- Compression temporarily perturbs the system away from equilibrium ($Q_p \neq K_p$) [1]
- The position of equilibrium shifts to restore the ratio back to K_p [1]
- Thus the composition changes, but the value of K_p remains strictly constant at 0.200 atm [1].

(e) [3 Marks]

- For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $Q_p = (2p_{\text{HI}})^2 / [(2p_{\text{H}_2})(2p_{\text{I}_2})] = 4(p_{\text{HI}})^2 / [4(p_{\text{H}_2})(p_{\text{I}_2})]$ [1]
- The factor of 4 cancels out completely in numerator and denominator: $Q_p = K_p$ [1]
- Because Q_p remains exactly equal to K_p upon compression, no shift in equilibrium occurs [1].

Question 33: Calculating K from Total Entropy for CaCO_3 (18 Marks)

(a) [2 Marks]

- $S^\circ_{\text{system}} = S^\circ(\text{CaO}) + S^\circ(\text{CO}_2) - S^\circ(\text{CaCO}_3) = 38.2 + 213.6 - 92.9$ [1]
- = $+158.9 \text{ J K}^{-1} \text{ mol}^{-1}$ [1].

(b) [5 Marks]

- $S_{\text{surr}} = -H^\circ / T = -(+178\,300) / 298 = -598.32 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $S_{\text{total}} = +158.9 + (-598.32) = -439.42 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $\ln K_p = S_{\text{total}} / R = -439.42 / 8.314 = -52.853$ [1]
- $K_p = \exp(-52.853) = 1.1 \times 10^{-23} \text{ atm}$ [2].

(c) [4 Marks]

- At 1200 K: $S_{\text{surr}} = -(+178\,300) / 1200 = -148.58 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $S_{\text{total}} = +158.9 + (-148.58) = +10.32 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $\ln K_p = +10.32 / 8.314 = +1.2413$ [1]
- $K_p = \exp(+1.2413) = 3.46 \text{ atm}$ (accept 3.4 to 3.5 atm) [1].

(d) [3 Marks]

- $K_p = p(\text{CO}_2)$ (solids omitted) [1]
- Since $K_p = 3.46 \text{ atm}$ at 1200 K, the equilibrium partial pressure $p(\text{CO}_2)$ is 3.46 atm [2].

(e) [4 Marks]

- Blowing air continuously sweeps away CO_2 gas, keeping $p(\text{CO}_2)$ effectively near zero [1]
- $Q_p = p(\text{CO}_2) < K_p$ at all times [1]
- The equilibrium is continuously driven to the right according to Le Chatelier's principle [1]
- 100% conversion is achieved without needing excessively high temperatures [1].

Question 34: *Level of Response — Le Chatelier vs $S_{\text{total}} = R \ln K$ (18 Marks)

(a) [3 Marks]

- Forward reaction is exothermic (releases heat) [1]
- By Le Chatelier, increasing temperature causes system to shift in endothermic direction (reverse) to absorb added heat [1]
- Yield of ammonia decreases [1].

*(b) [6 Marks — Level of Response]

- Level 3 (5–6 marks): Comprehensive comparison between Le Chatelier and thermodynamics. Explains that Le Chatelier predicts direction of shift phenomenologically, but $S_{\text{total}} = R \ln K$ provides quantitative, fundamental explanation. Explains that $S_{\text{surr}} = -H/T$. For exothermic reactions ($H < 0$), higher T makes positive S_{surr} smaller, reducing S_{total} , which reduces K (fewer products). For endothermic ($H > 0$), higher T makes negative S_{surr} less negative, increasing S_{total} , which increases K . Rigorous catalyst explanation: A catalyst lowers activation energy for BOTH forward and reverse pathways equally, speeding up both rates by identical factors. H and S_{system} are state functions that depend ONLY on reactant and product identities, not mechanism. Thus S_{surr} and S_{total} are completely unaffected by catalyst, meaning K and equilibrium yield cannot change.
- Level 2 (3–4 marks): Correctly links $S_{\text{surroundings}}$ to temperature and explains catalyst effect on activation energy vs state functions with minor gaps.
- Level 1 (1–2 marks): Descriptive restatement of Le Chatelier with superficial mention of entropy.

(c) [4 Marks]

- Observation: Mixture turns darker brown [1]
- Decomposition is endothermic ($H = +57 \text{ kJ mol}^{-1}$) [1]
- Increasing T makes negative S_{surr} less negative, increasing S_{total} and K_p [1]
- Equilibrium shifts right, producing more brown NO_2 gas [1].

(d) [5 Marks]

- Thermodynamic penalty: At 450 °C, K_p is small because reaction is exothermic, limiting single-pass yield to ~15% [2]
- Kinetic requirement: At room temperature, rate is virtually zero because breaking the NN triple bond has a huge activation energy (945 kJ mol^{-1}) [2]
- Economic compromise: 450 °C with iron catalyst gives an acceptable reaction velocity to reach equilibrium in seconds; unreacted gases are recycled [1].

Question 35: Brønsted–Lowry Equilibria & Non–Aqueous Systems (18 Marks)

(a) [4 Marks]

- (i) Pair 1: HNO_2 (acid) / NO_2^- (base) [1]; Pair 2: H_3O^+ (acid) / H_2O (base) [1]
- (ii) Pair 1: H_2SO_4 (acid) / HSO_4^- (base) [1]; Pair 2: NH_4^+ (acid) / NH_3 (base) [1].

(b) [4 Marks]

- (i) Acid: $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ [2]
- (ii) Base: $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) + \text{OH}^-(\text{aq})$ [2].

(c) [4 Marks]

- (i) Autoionisation: $2\text{NH}_3(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{am}) + \text{NH}_2^-(\text{am})$ [2]
- (ii) Neutralisation: $\text{NH}_4^+ + \text{NH}_2^- \rightarrow 2\text{NH}_3$ [2].

(d) [3 Marks]

- Nitric acid, HNO_3 , acts as a Brønsted-Lowry base [1]
- H_2SO_4 is a much stronger acid than HNO_3 and forces HNO_3 to accept a proton on its $-\text{OH}$ group (forming H_2NO_3^+) [1]
- H_2NO_3^+ subsequently loses water to generate the electrophilic nitronium ion, NO_2^+ [1].

(e) [3 Marks]

- In pure water, $[\text{H}^+] = [\text{OH}^-] = \sqrt{K_w} = \sqrt{5.13 \times 10^{-13}} = 7.16 \times 10^{-7} \text{ mol dm}^{-3}$ [1]
- $\text{pH} = -\log[\text{H}^+] = -\log(7.16 \times 10^{-7}) = 6.14$ [1]
- Water is strictly neutral because $[\text{H}^+]$ equals $[\text{OH}^-]$; neutral pH is 6.14 at 100 °C because dissociation is endothermic [1].

Question 36: Ocean Acidification & Coral Calcification (15 Marks)

(a) [4 Marks]

- Rising atmospheric $p(\text{CO}_2)$ shifts Equilibrium 1 to the right (more CO_2 dissolves) [1]
- Increased $[\text{CO}_2(\text{aq})]$ shifts Equilibrium 2 right to form more carbonic acid, H_2CO_3 [1]
- Increased $[\text{H}_2\text{CO}_3]$ shifts Equilibrium 3 right, releasing more H_3O^+ ions [1]
- $[\text{H}_3\text{O}^+]$ increases, causing seawater pH to fall (ocean acidification) [1].

(b) [3 Marks]

- Added H_3O^+ ions react with basic carbonate ions: $\text{H}_3\text{O}^+ + \text{CO}_3^{2-} \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$ [1]
- This neutralisation consumes carbonate ions, shifting Equilibrium 4 to the left [1]
- Consequently, the concentration of free carbonate ions $[\text{CO}_3^{2-}]$ decreases substantially [1].

(c) [5 Marks]

- (i) $= (0.0102 \times 2.40 \times 10^{-4}) / (6.0 \times 10^{-9}) = (2.448 \times 10^{-6}) / (6.0 \times 10^{-9}) = 408$ (strongly supersaturated, $\gg 1$) [2]
- (ii) New $= (0.0102 \times 5.0 \times 10^{-7}) / (6.0 \times 10^{-9}) = (5.10 \times 10^{-9}) / (6.0 \times 10^{-9}) = 0.85$ [2]
- Since < 1 , seawater is undersaturated with respect to aragonite; existing coral skeletons spontaneously dissolve [1].

(d) [3 Marks]

- Ecological: Loss of marine biodiversity and habitats for 25% of all fish species supported by reefs [1]
- Economic: Collapse of commercial coastal fisheries and loss of ecotourism revenue [2].

Question 37: Haber-Bosch Thermodynamic Loop Optimisation (15 Marks)

(a) [4 Marks]

- $S_{\text{surr}} = -(-92\,200) / 298 = +309.40 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $S_{\text{total}} = -198.8 + 309.40 = +110.60 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $\ln K_p = 110.60 / 8.314 = 13.303$ [1]
- $K_p = \exp(13.303) = 6.0 \times 10^5 \text{ atm}^{-2}$ (accept 5.8×10^5 to 6.8×10^5) [1].

(b) [4 Marks]

- At 723 K: $S_{\text{surr}} = -(-92\,200) / 723 = +127.52 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $S_{\text{total}} = -198.8 + 127.52 = -71.28 \text{ J K}^{-1} \text{ mol}^{-1}$ [1]
- $\ln K_p = -71.28 / 8.314 = -8.5735$ [1]
- $K_p = \exp(-8.5735) = 1.9 \times 10^{-4} \text{ atm}^{-2}$ (accept 1.4×10^{-4} to 2.0×10^{-4}) [1].

(c) [4 Marks]

- K_p at 723 K is roughly 10^9 times smaller than at 298 K [1]
- At 298 K, reaction rate is imperceptibly slow due to high activation energy of N-N bond [1]
- High temperature (450 °C) ensures rapid kinetic collision rate to achieve equilibrium in seconds [1]
- High pressure (200 atm) partially compensates for the thermodynamic penalty by shifting equilibrium right [1].

(d) [3 Marks]

- Gases leaving reactor are cooled under pressure, liquefying ammonia (bp -33 °C) while N_2 and H_2 remain gases [1]

- Liquid ammonia is tapped off [1]
 - Unreacted N_2 and H_2 are recycled continuously back into the catalyst bed, achieving overall 98% yield [1].
-